Title: Discovering Acids and Bases

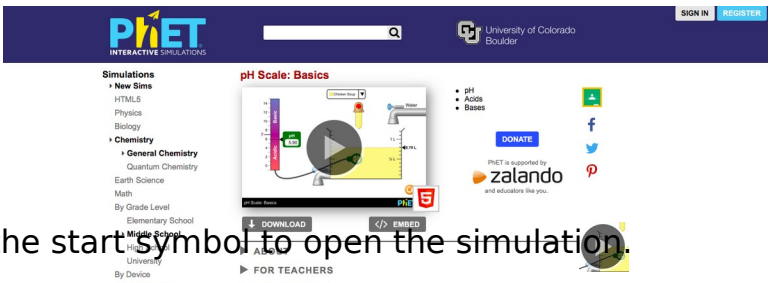
WHAT ARE ACIDS AND BASES?
Acids and ***bases*** are two kinds of special chemicals. Almost all liquids are either ***acids*** or ***bases***, which is determined by the type of ions in them. An ***acid*** has many hydrogen ions, and **splits apart** in water to release these hydrogen ions. On the contrary, ***bases*** have hydroxide ions that **combine** with hydrogen ions. Scientists us a pH scale to determine whether a liquid is an ***acid*** or a ***base***. PH refers to hydrogen ion concentration, which is measured by a scale. The scale goes from 0-14, where liquids having a pH level of 0-7 are ***acids***, and 7-14 are ***bases***. Any substance with a pH level of 7 is considered neutral.

IMPORTANT QUESTIONS:
What substances are ***acids/bases***?
How are all ***acids/bases*** similar?
What happens to the pH level of substances when it is diluted?

INSTRUCTIONS:
Using the basic pH scale simulation on the PhET website, we will explore and answer the important questions of this investigation. We will gain experience using a virtual pH scale, and test the pH levels of different substances in order to determine them as an ***acid*** or ***base***. Based on our results, we will analyze how the discovered ***acids*** are similar, and discovered ***bases*** are similar. We will use our discoveries to explore and understand how pH levels of substances are affected when diluted.

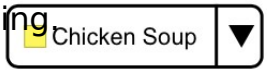
<u>Introduction-Explore</u>
1. Copy and paste this website you're your Internet browser: https://phet.colorado.edu/en/simulation/ph-scale-basics

Your page should look like this:



2. Click on the start symbol to open the simulation.

3. Press the drop down at the top of the screen to view the substances you will be testing.



How many different liquids can you test with this simulation?

_____

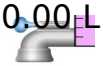
4. Choose a liquid and use the dropper to fill the beaker to 1/2L.



5. Fill the beaker to 1L by clicking the water faucet continuously until the liquid level reaches the 1L mark.



6. Empty your beaker by clicking the bottom faucet continuously until the measurement arrow reads 0.00 L.



Prediction

Predict which substances in the chart will be determined as ***acids*** and ***bases*** on the pH scale.

Substance	Acid or Base
Drain Cleaner	
Hand Soap	
Blood	
Milk	
Chicken Soup	
Coffee	
Orange Juice	
Vomit	
Water	

Explain how/why you categorized the substances the way you did. What characteristics of the substances helped you predict whether the substance was an ***acid*** or ***base***?

_____

_____

_____

_____

Investigation

1. Set the substance to drain cleaner.
2. Press the dropper and fill the beaker to ½ L.
3. Place the pH scale sensor in the beaker so that the center of the sensor is touching the liquid in the beaker.
4. Observe the pH reading on the scale.

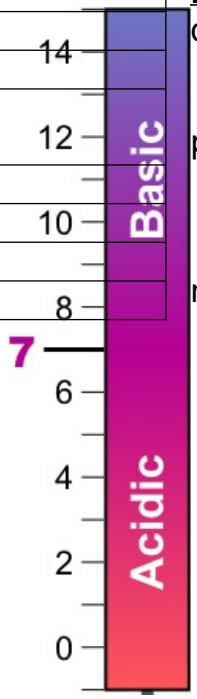


5. Record the substance's pH level and whether it is an acid or a base in the table below.
6. Repeat these steps with each substance listed in the table.

Substance	pH Level	Acid or Base
Drain Cleaner		
Hand Soap		
Blood		
Milk		
Chicken Soup		
Coffee		
Orange Juice		
Vomit		
Water		

Analyzing Data

Analyze your data, and place each substance in the proper place on the pH scale.



Which substance is ***acidic***?

_____

Which substance is most ***basic***?

_____

Which substance is considered ***neutral***?

_____

Which substance surprised you the most and why?

_____

What do you notice about most of the substances that are considered ***acids***? What do they have in common?

_____

_____

_____

_____

_____

What do you notice about most of the substances that are considered ***bases***? What do they have in common?

_____
_____
_____
_____
_____

Which substances were closest to ***neutrality***? What do they have in common?

_____
_____
_____
_____
_____

Other substances in the simulation that were not tested are **spit, battery acid** and **soda pop**. Based on your data and observations of ***acids*** and ***bases***, which category would you place these substances in and why? Check your answer using the simulation.

_____
_____
_____
_____

Why do you think water is considered a ***neutral*** substance?

_____
_____
_____
_____

<u>Extension-</u>What happens to the pH level of a substance when it is diluted?

Using what you know about ***acids*** and ***bases***, **predict** what will happen to the pH level when water is mixed with each substance?

I think the pH level will increase/decrease/stay the same/vary (circle) when diluted because…

_____
_____

Steps:

1. Based on your previous investigation, write whether each substance is an ***acid*** or ***base*** in the chart below.
2. Set the substance to drain cleaner.
3. Press the dropper and fill the beaker to ½ L.

4. Measure and record the pH level of the substance.
5. Keeping the pH sensor in your beaker add water until the beaker is filled to 1.20 L.
6. Record the substance's pH level after it has been diluted.
7. Empty your beaker by clicking on the bottom faucet, or the orange refresh symbol.
8. Repeat these steps with each substance listed in the table.

Making Observations and Analyzing Data:

1. What patterns do you notice in your data? Based on your data, explain what happens to the pH level of ***acids*** and ***bases*** when they are diluted.

_____

Substance	**Acid or Base**	**pH at 1.20 L**	**pH at 1.20 L**
Drain Cleaner			
Hand Soap			
Blood			
Spit			
Milk			
Chicken Soup			
Coffee			
Orange Juice			
Soda Pop			
Vomit			
Battery Acid			

_____

_____
_____
_____

2. Besides pH level, what other changes did you notice when each substance was diluted with water?

_____
_____
_____
_____
_____

3. By adding water to each substance, the amount of solute is decreased. By adding water to the solution, the amount of H+ ions also decreases. In relation to your data, how do you think this affects pH levels?

_____
_____
_____
_____
_____
_____